THE ECHO JOURNAL

A Guide to Transthoracic Imaging Windows

Overview

Transthoracic echocardiography (TTE) uses a phased array probe to transmit sound waves into the body and generate an image. Inside the probe, an electrical current vibrates piezoelectric crystals, which send sound waves into the body. These crystals also detect the returning waves and send them to scan converters in the system. The resulting reflections, or “echoes,” are then used to create a diagnostic image.

The speed of sound propagation is determined by the medium through which it travels, with sound waves moving slowest through air, gas, and adipose tissue. To optimize imaging, the probe is tilted, rotated, slid, rocked, or angled within each acoustic window described below, with the resulting images categorized based on the axis plane.

Objectives

- Key terms and definitions
- Learn techniques to capture and improve standard TTE images
- Understand normal cardiac anatomy seen in each view
- Master probe placement, patient positioning, and areas of interest for each view

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Glossary

Left Heart	Right Heart
Left ventricle (LV)	Right ventricle (RV)
Left atrium (LA)	Right atrium (RA)
Mitral valve (MV)	Tricuspid valve (TV)
Anterior mitral leaflet (AMVL)	Septal leaflet (S)
Posterior mitral leaflet (PMVL)	Posterior leaflet (P)
Left ventricular outflow tract (LVOT)	Anterior leaflet (A)
Aortic valve (AV)	Pulmonic valve (PV)
Right coronary cusp	Right ventricular inflow tract (RVIT)
Non coronary cusp	Right ventricular outflow tract (RVOT
Left coronary cusp	Moderator band
Aortic root (AoRoot)	Right ventricular free wall
Ascending Aorta (AscAo)	Crista Terminalis
Interventricular septum (IVS)	Eustachian Valve (EV)
Interatrial septum (IAS)	Posteromedial papilllary muscle in LV (PMPap)
Innominate artery (IA)	Anterolateral papillary muscle in LV (AMPap)
Left common carotid artery (LCCA)	Coronary Sinus
Left subclavian artery (LSA)	Anterior papillary muscle in RV (APM)
Descending Aorta (DAo)	Posterior papillary muscle in RV (PPM)

Two-Dimensional (2D): visualize structures in real time by taking a cross-sectional slice of the heart to see ventricles, atria, and valves. This is limited by patient body habitus, air lung interference, and the operator of the machine.

M-Mode: also known as “motion mode” is a single scan line to display a graph of motion over time. It has excellent temporal resolution and is best for timing of events when perpendicular to the structure.

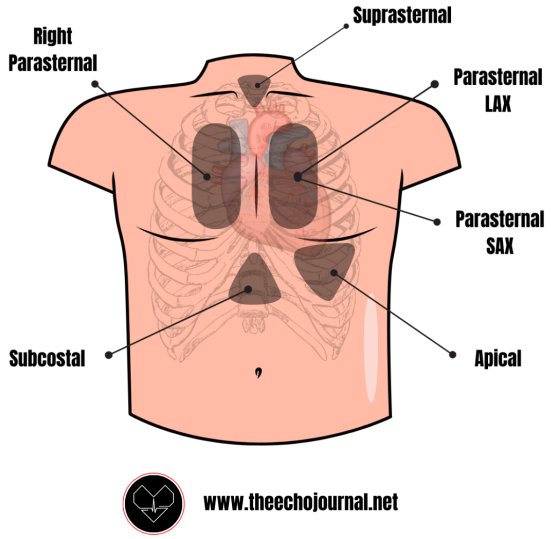
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Spectral Doppler: displays a blood flow velocity graph as a waveform and is more precise when the ultrasound beam is parallel to the direction of flow. The two types used are pulsed wave Doppler and continuous wave Doppler.

Color Doppler: used over a 2D image to assess blood flow and direction for valve competency, interatrial shunts, or interventricular shunts. Quality is dependent on red blood cells being parallel to the transducer and a narrower color box will improve the image.

5 Standard Imaging Windows

- Parasternal Long Axis View (PLAX)
- Parasternal Short Axis (PSAX)
- Apical (A)
- Subcostal (SC)
- Suprasternal (SSN)



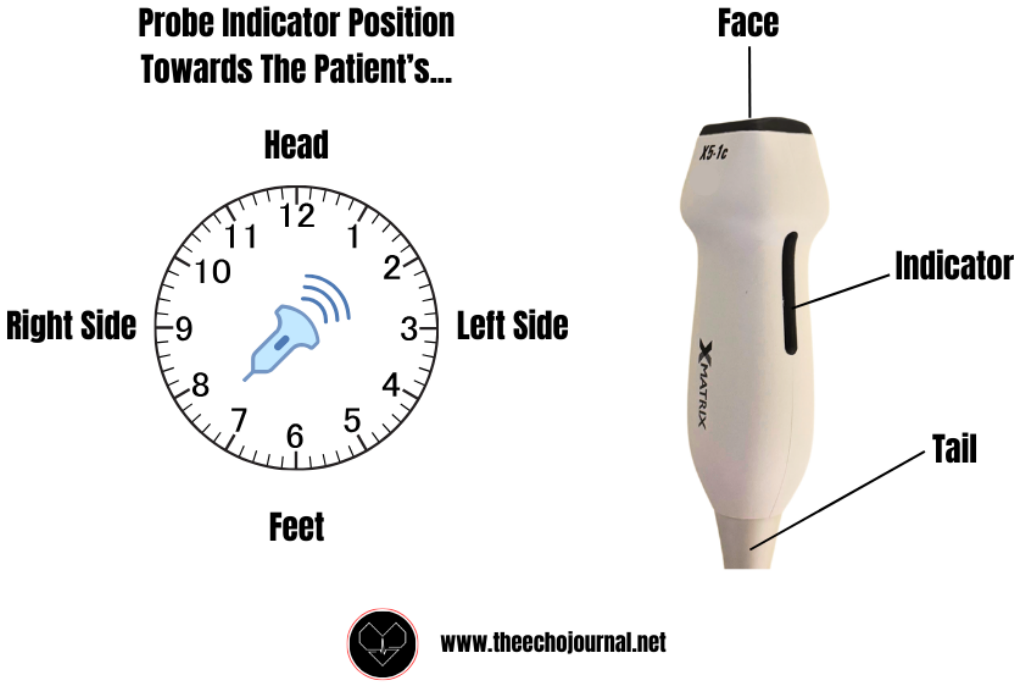
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Key Terms

Protocol: a systematic approach to acquiring standard images when performing a complete transthoracic echocardiogram (TTE). An adult TTE begins with the patient positioned in the left lateral decubitus position – or left side. This moves the heart closer to the chest wall and the ultrasound probe, which enhances image quality. For each standard imaging window listed below, a set of recommended views and measurements should be taken. While this guide is universally applicable, measurement values should align with your local guidelines.

Window: a specific anatomical area on the body where ultrasound waves can easily pass through.

Indicator: a marking on the side of the ultrasound probe that corresponds to a marker on the screen. It helps identify the probe's positioning when rotated to locate standard imaging windows.



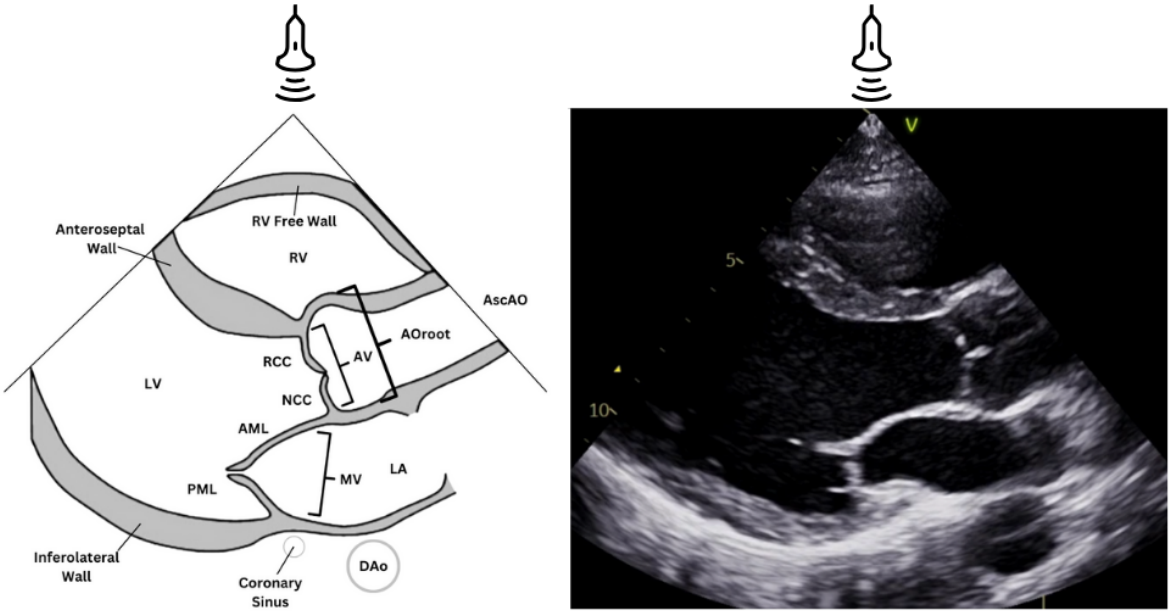
On-axis: a term used to describe if an image is in the proper window for accurate assessment.

Foreshortened: the probe is improperly positioned and the image is not aligned perpendicular to the area of interest; measurements may be inaccurate as a result.

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Parasternal Long Axis View (PLAX)

Patient positioning: have the patient lie in a left lateral decubitus position (left side onto the hip) with the right arm down by the side and the left arm behind the head. The patient should be relaxed and breathing normally.



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Probe placement: place the probe to the left of the patient's sternum perpendicular to the chest wall. This is typically at the 3rd or 4th intercostal space. The indicator should be turned towards the patient's right shoulder so that the beam is perpendicular to the left ventricle cavity.

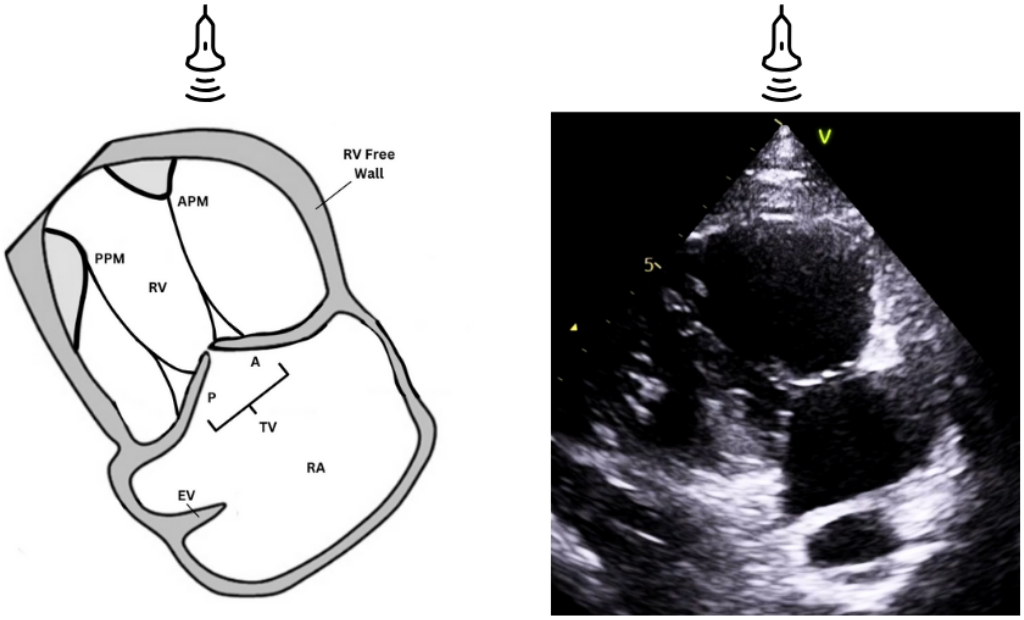
Quick tips

- Start higher up on the chest and work your way down instead of starting lower and working your way up to obtain an on-axis image.
- Observe how small rotations and tilts can open up the left ventricle, mitral, and aortic valves.

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Right Ventricular Inflow Tract View (RVIT)

Patient positioning: have the patient lie in a left lateral decubitus position (left side onto the hip) with the right arm down by the side and the left arm behind the head. The patient should be relaxed and breathing normally.



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Probe placement: from the parasternal long axis view, bring the tail of the probe towards the patient's head and angle the face of the transducer towards the right hip. This simple tilting motion will bring in the right ventricular inflow tract view – displaying the right atrium, tricuspid valve, and right ventricle. You may need to slide the probe laterally.

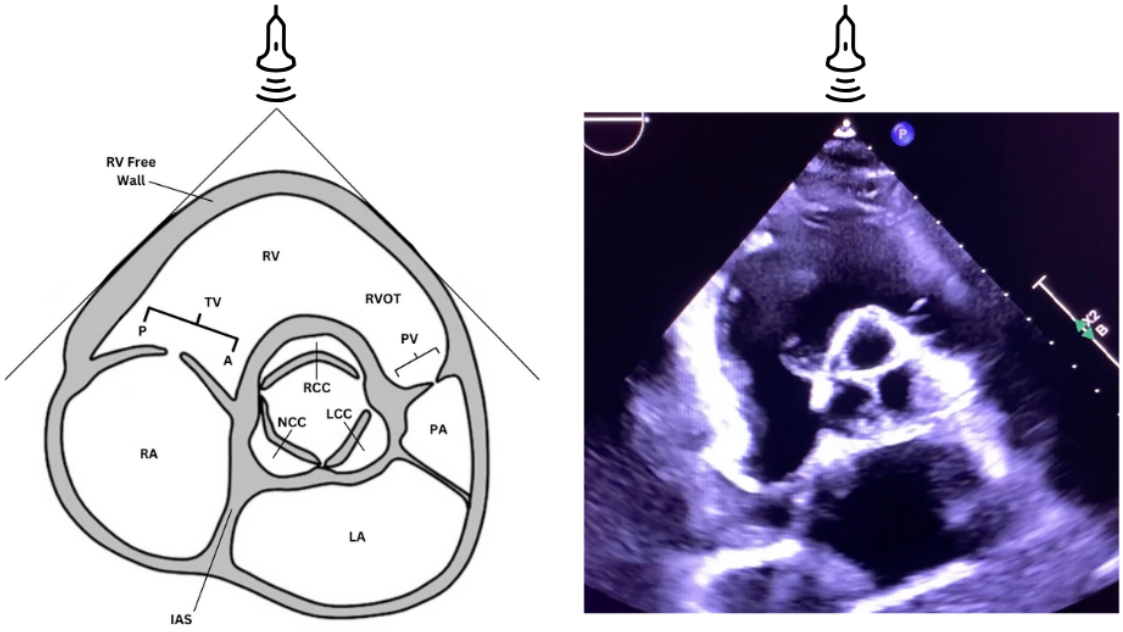
Areas of Interest:

- The first look at the tricuspid valve at the beginning of the protocol.
- Evaluates the highest velocity of tricuspid regurgitation using CW Doppler using the simplified Bernoulli principle.
- If the interventricular septum is *not* seen then it the anterior and posterior leaflets in view, if the interventricular septum *is* seen then the anterior and septal leaflets are in view.

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Parasternal Short Axis View (PSAX) Aortic Valve Level

Patient positioning: the patient will stay in the left lateral decubitus position with the right arm down by the side and the left hand behind the head. The patient should be relaxed and breathing normally.



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Probe placement: this image is in the same location as the PLAX view but the indicator is at the 12 o'clock position towards the patient's head - this is a short axis view of the heart. Use the PLAX view as a home base to reset if you lose your image.

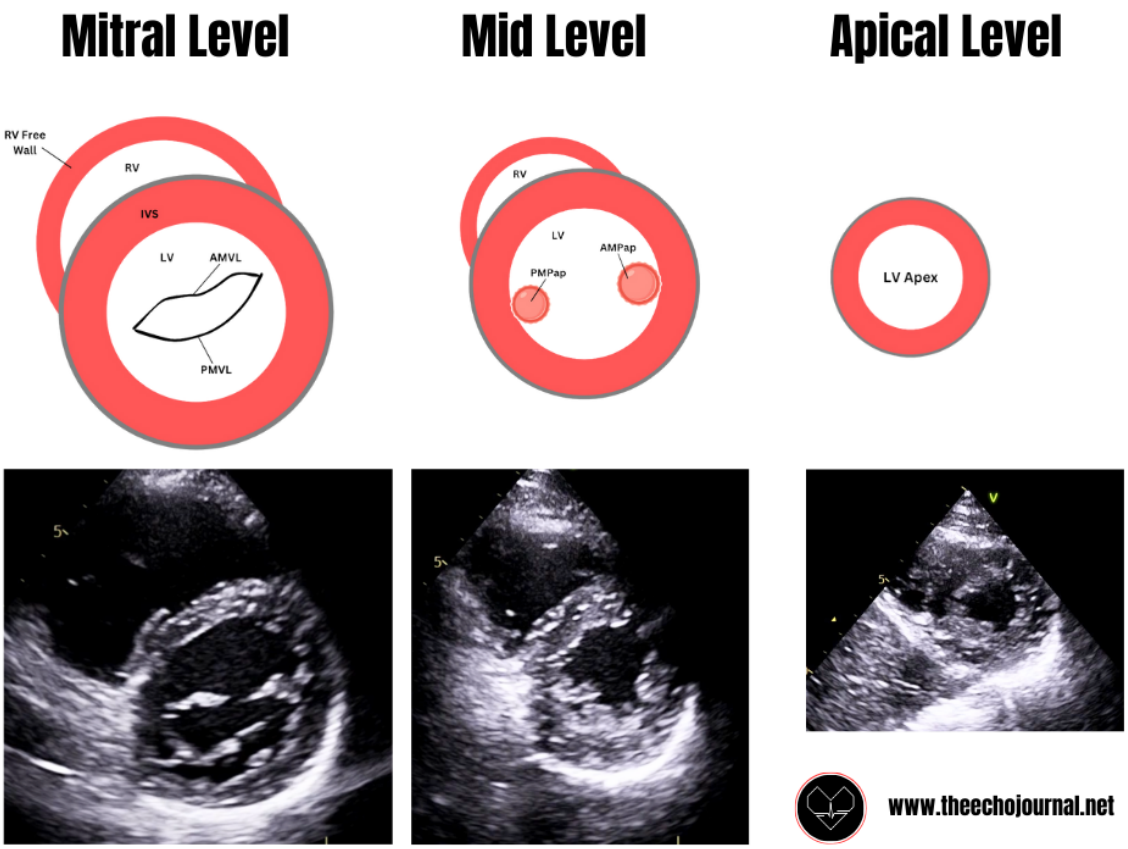
Areas of Interest

- The 3 aortic valve leaflets are used as a landmark in the middle of the screen and will look like a Mercedes Benz logo.
- Tilting the probe helps better visualize tricuspid regurgitation (TR) with color Doppler.
- Continuous wave Doppler is helpful in this view to get the highest velocity of TR and estimate right ventricular systolic pressures (RVSP).

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Parasternal Short Axis View (PSAX) Left Ventricle Level

The positioning, probe rotation, and indicator alignment will be the same as the parasternal short axis (PSAX) view at the aortic level, except the probe is tilted. From the PSAX aortic level, tilt the probe so that the face is angled toward the patient's left nipple.



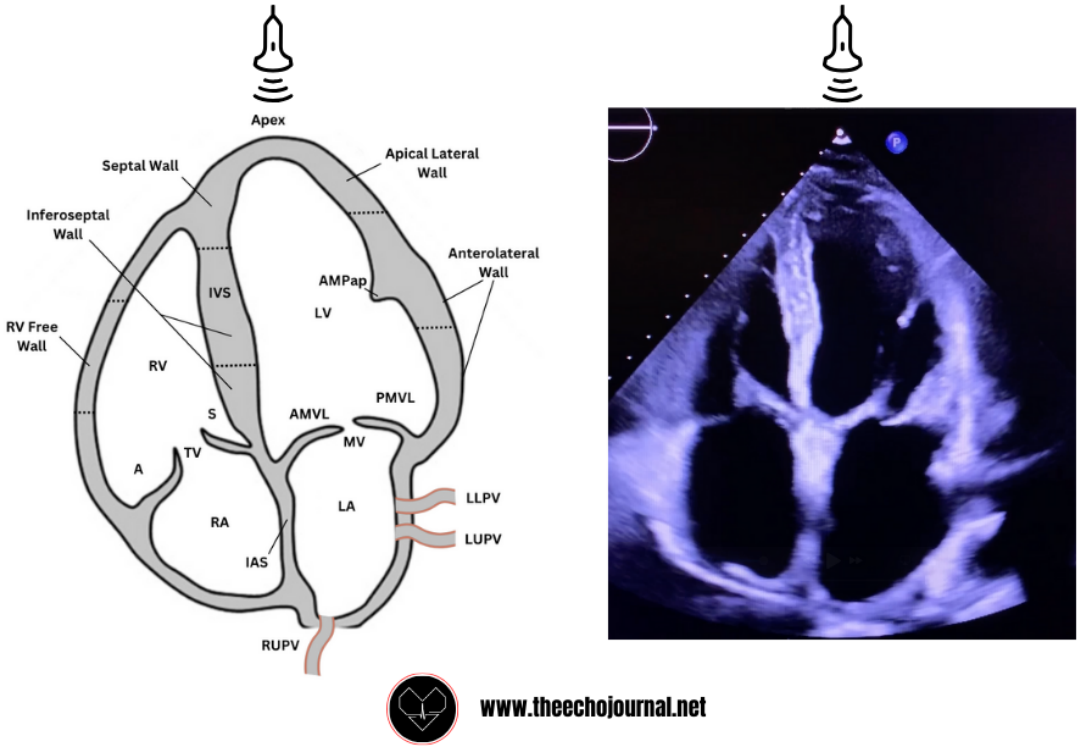
If you get stuck or lose your reference, return to the "home base" image – the parasternal long axis view, with the indicator pointing toward the patient's right shoulder. From there, rotate the indicator to the 12 o'clock position to get the parasternal short axis view showing the aortic valve. Without moving the probe, simply angle the face of the transducer slightly towards the patient's left nipple.

As you adjust the probe, the mitral valve should come into view, with the anterior leaflet positioned on top and the posterior leaflet below. You may need to make minor rotations and tilts to fine-tune the image. Sliding the probe laterally may be helpful here as you slice through the ventricle at the basal, middle, and apical levels.

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Apical 4 Chamber View

Patient positioning: the patient will stay in the left lateral decubitus position while breathing normally. Lung interference can negatively affect image quality – experiment with a small, medium, or large breath in to expand the lungs and bring the heart into the best acoustic window.



Probe placement: the probe will be placed at the apex of the heart, typically at the 4th or 5th intercostal space, underneath the left breast or left pectoral muscle. The indicator will be towards the bed at about the 3 o'clock position.

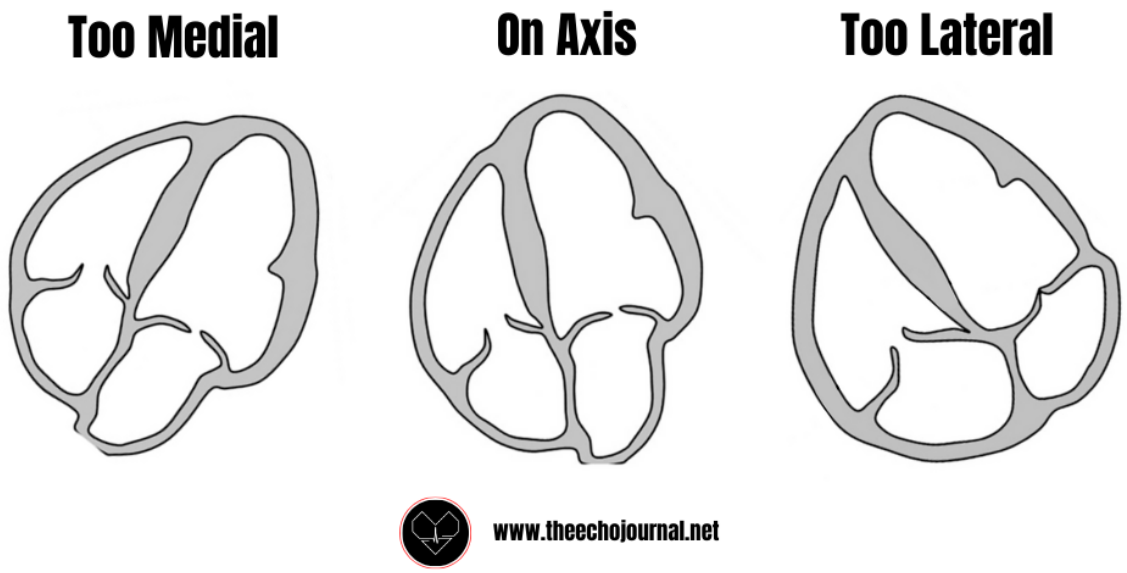
Areas of Interest

- The interventricular septum and interatrial septum should be in the middle of the screen oriented vertically. The four chambers and mitral and tricuspid valves should be visible.
- Size and function of the ventricles can be evaluated in this view.
- Mitral valve function is thoroughly assessed in this view using 2D, color, and spectral Doppler.

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Avoid Foreshortening

A foreshortened apical image can result in inaccurate assessment of left ventricular wall motion contractility. If the apex is bouncing downwards, the probe needs to slide into a more lateral position. See the following graphic and note the interventricular septum location.



Quick Tips

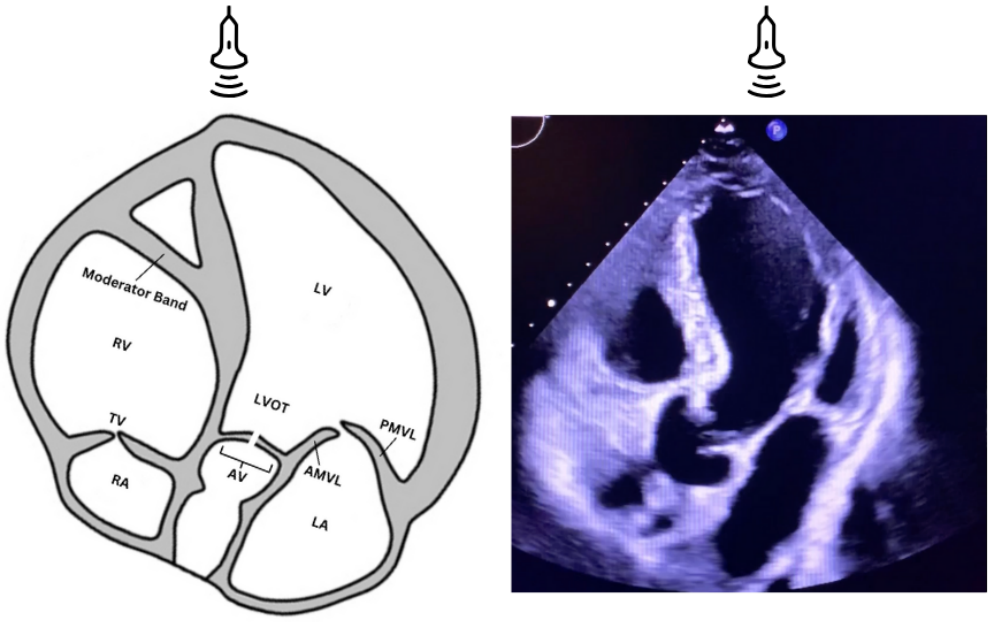
- A medial apical 4 chamber view is best for getting parallel to the highest tricuspid regurgitation.
- An on axis apical 4 chamber view is used for the most accurate assessment of left ventricular regional wall motion contractility.
- A lateral view is best for linear measurements of the right ventricular cavity and free wall contractility.

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Apical 5 Chamber View

Patient positioning: the patient will stay in the left lateral decubitus position while breathing normally. Tilting the patient's right shoulder back towards a more supine position can make more room for the probe as the objective is to get parallel to transaortic flow.

Probe placement: from the apical 4 chamber image, with the probe indicator at the 3 o'clock position - rotate clockwise to the 4 o'clock position. This will open up the left ventricular outflow tract and the aortic valve will come into view.



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Areas of Interest

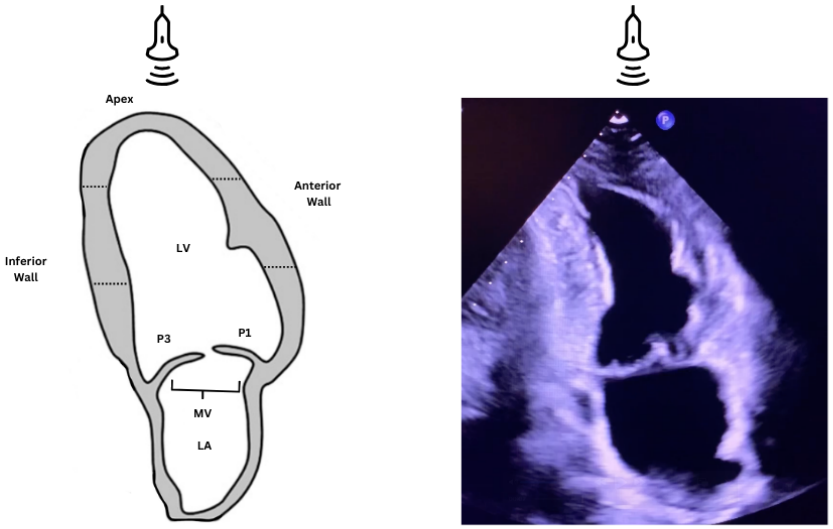
- Spectral Doppler is used routinely to assess the velocities through the aortic valve – this is particularly important in cases of aortic stenosis.
- Getting parallel to transaortic flow is the objective to achieve the highest velocity with CW Doppler.
- The continuity equation is the key formula here as it is used to calculate the aortic valve area and the stroke volume, or volume of blood pumped out with each beat.

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Apical 2 Chamber View

Patient Positioning: the patient will stay in the left lateral decubitus position while breathing normally. If you have difficulty viewing the anterior wall, roll the patient more onto their left side almost bringing the right side of their chest over the probe. Experimenting with a small, medium, or large breath in can also be helpful here.

Probe Placement: from the apical 4-chamber image rotate the indicator to the 12 o'clock position



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Areas of Interest

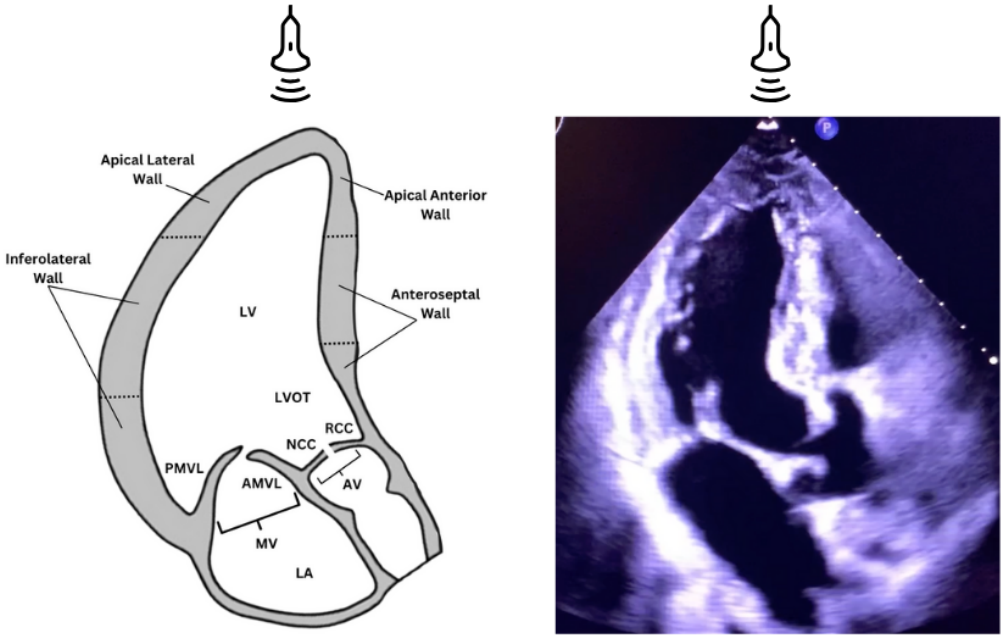
- This is an orthogonal (90 degree) view of the left ventricle when compared to the apical 4 chamber view.
- Size and function of the left ventricle are assessed here.
- The anterior left ventricular wall is fed oxygenated blood by the left anterior descending (widow maker) and visualization is crucial.
- Mitral valve function is also be assessed with 2D, color, and spectral Doppler.

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Apical 3 Chamber View

Patient positioning: the patient will stay in the left lateral decubitus position while breathing normally. Tilting the patient and experimenting with small, medium, or large breaths in may help optimize.

Probe placement: from the apical 2 chamber image rotate the probe counterclockwise to the 10 o'clock position. The left ventricular outflow tract and the aortic valve will come into view next to the posterior mitral leaflet.



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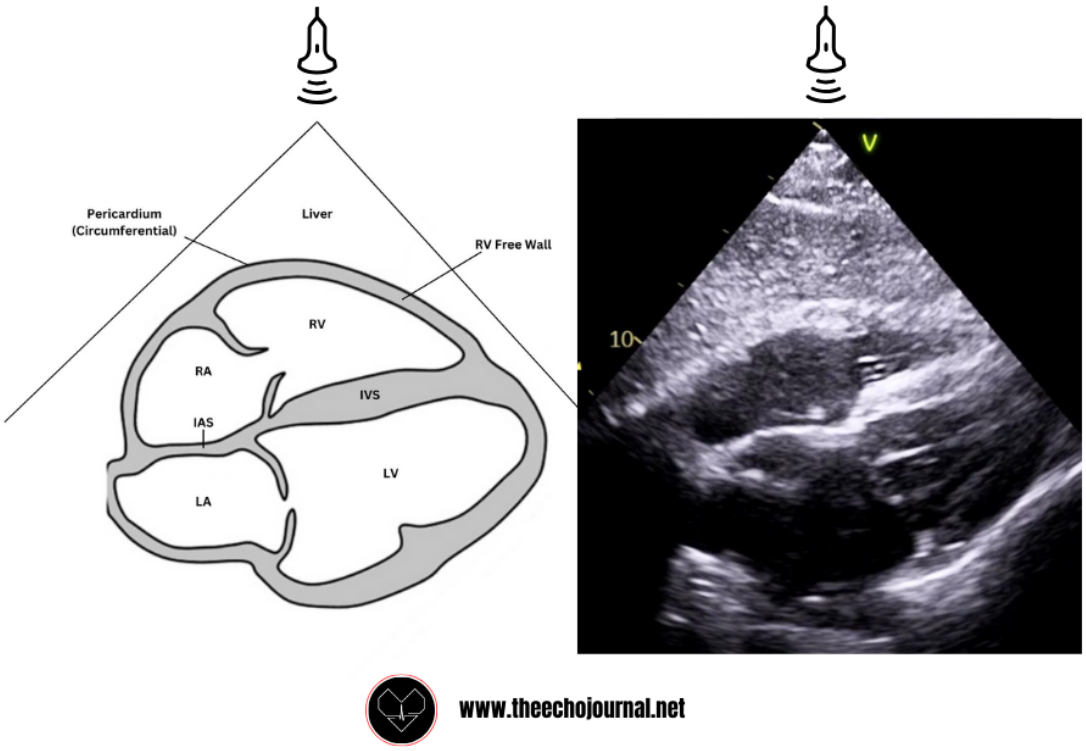
Areas of Interest

- This view brings the sonographer parallel to the flow through the aortic valve similar to the 5 chamber view.
- Size and function of the left ventricle are assessed here.
- Mitral valve function is thoroughly assessed in this view using 2D, color, and spectral Doppler.

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Subcostal 4 Chamber view

Patient positioning: the patient will lie flat on their back in the supine position with the knees bent to relax the abdominal muscles. A small, medium, or large breath in might also be needed to open the diaphragm to visualize the heart.



Probe placement: the probe will be placed at the subxiphoid location with the indicator at the 3 o'clock position just beneath the xiphoid process. The probe will need to be flat against the patient's abdomen with an overhand probe grip.

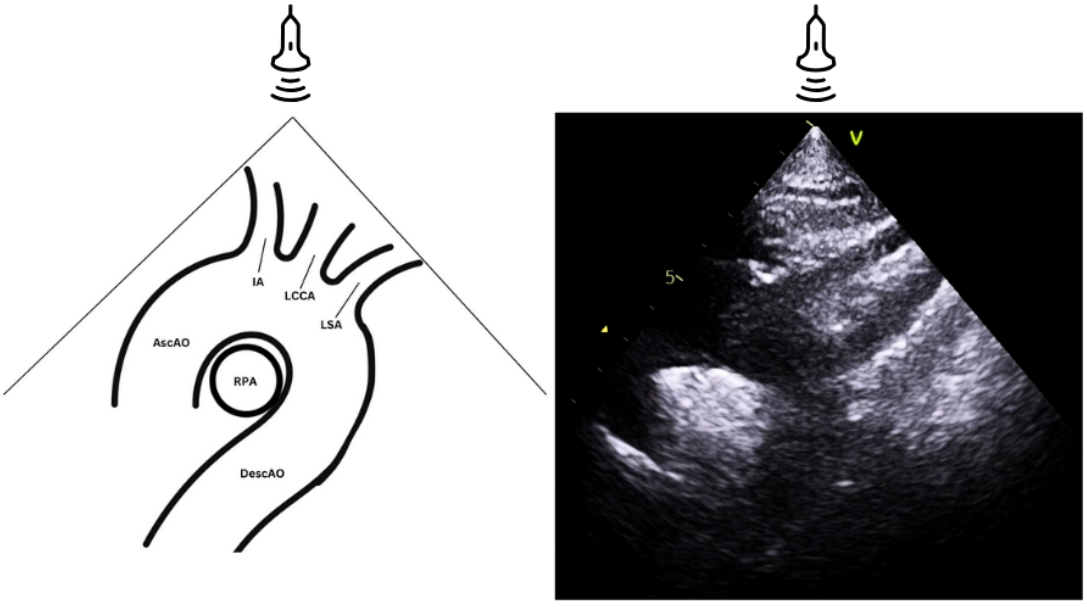
Areas of Interest

- Atrial septal defect (ASD) and patent foramen ovale (PFO) are best seen in this view.
- Useful in the evaluation of RV free wall and RA wall inversion in cases of significant pericardial effusion when looking for cardiac tamponade – a critical echo finding.
- RV wall thickness best measured in this view.

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Suprasternal Notch View

Patient positioning: the patient will lie flat on their back in the supine position with the chin tilted slightly upward. It can be beneficial to have the patient turn their head to the left or have them exhale any excess air.



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Probe placement: place the probe in the suprasternal notch with the indicator at the 12 o'clock or 1 o'clock position and angle toward the plane that cuts through the right nipple and the tip of the left scapula.

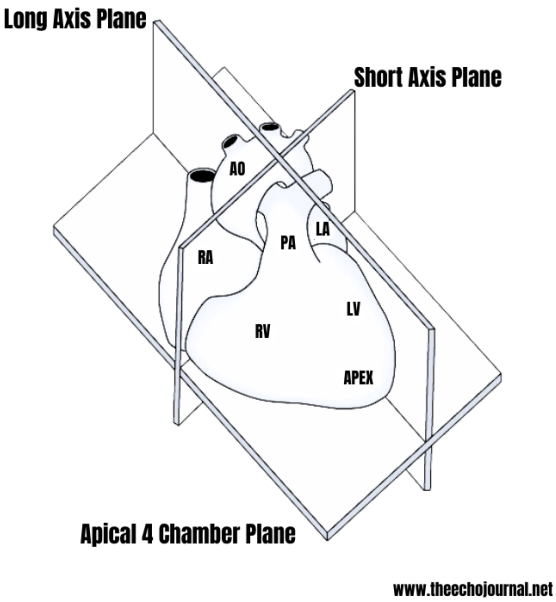
Areas of Interest

- The ascending, transverse, and descending aorta are seen.
- This image is helpful in cases of significant aortic regurgitation looking for flow reversal in the descending aorta.
- Accelerated flow in the ascending aorta can also be assessed in this view in cases of aortic stenosis.

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